

# Assumptions Made Visible: What Building a Macroeconomic Scenario Simulator Taught Me About Empirical Modelling

*Data Lab Research Note II — Elasticity Rules, Shock Transmission, and the Boundary Between  
Illustration and Estimation*

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## Introduction — The Temptation of the Black Box

When I first designed the interactive chart for the Macroeconomic Data Lab, I wanted to go beyond plotting historical data from 2019 to 2023 and explore what happens when an economy is hit by a new policy-rate hike or a supply-side price shock. In student projects — and often in financial commentary — there is a strong temptation to dress up a simple formula in impressive econometric jargon, calling a spreadsheet calculation a “Structural Vector Autoregression” or an “algorithmic forecast.”

As I read more deeply into how empirical macroeconomics actually works, I realised that pretending a simple rule is an estimated model gets the scientific mindset exactly backwards. A model is useful not because its label sounds sophisticated, but because its assumptions are transparent enough to be questioned. That realisation led me to redesign the Data Lab’s 2024–2025 extension as an explicitly illustrative Scenario Simulator — a tool that makes my elasticity assumptions visible so that anyone using the sliders can see both how textbook transmission channels work in theory and why fixed-parameter rules fall short of real-world econometrics.

## Section I — How the Scenario Simulator Is Constructed

The simulator extends the 2023 historical baseline into two hypothetical years (2024 and 2025) using two user-controlled sliders: a monetary policy-rate shock (measured in basis points, from 0 to +500 bps) and a cost-push supply-price shock (measured as a percentage increase, from 0% to +20%). To reflect the fact that monetary and supply shocks do not hit an economy all at once, the code applies a lag multiplier of 0.5 in the first year (2024) and 1.0 in the second year (2025). Each of the five economies is assigned three explicit sensitivity weights corresponding to how textbook open-economy macroeconomics — and my own reading on emerging markets — suggests they respond:

| Economy Cohort | Code | GDP Rate Elasticity | GDP Supply Weight | CPI Rate Elasticity |
|----------------|------|---------------------|-------------------|---------------------|
| Turkey         | TUR  | -0.22               | -0.65             | -0.85               |
| United States  | USA  | -0.35               | -0.35             | -0.45               |
| Germany        | DEU  | -0.28               | -0.55             | -0.38               |

|                |     |       |       |       |
|----------------|-----|-------|-------|-------|
| United Kingdom | GBR | -0.32 | -0.45 | -0.40 |
| Japan          | JPN | -0.08 | -0.40 | -0.12 |

*Table 1: Hand-set elasticity weights used in the Data Lab 2024–2025 Scenario Simulator (illustrative parameters, not estimated coefficients).*

Why did I choose these relative weights for the illustration? In the simulator, Germany (-0.55) and Turkey (-0.65) are given stronger negative output sensitivity to supply-price shocks than the United States (-0.35), reflecting their higher structural reliance on imported energy and intermediate inputs. Similarly, when a policy-rate shock is applied under the CPI view, Turkey is assigned a larger nominal disinflationary weight (-0.85) only because its baseline inflation rate (53.9% in 2023) is an order of magnitude higher than Japan’s (3.3%) or the US’s (4.1%) — meaning a multi-percentage-point rate adjustment moves a much larger nominal number, even when the underlying transmission channel is attenuated.

## Section II — Three Reasons a Fixed-Elasticity Rule Is Not an Econometric Model

Building this simulator myself was the fastest way to understand its limitations. There are three fundamental reasons why the 2024–2025 lines on the chart are a sensitivity exercise rather than a forecast or a causal estimate.

First, the parameters in Table 1 are assumed, not estimated from data. In my working paper on inflation persistence (WP-01), estimating even a single lagged inflation coefficient ( $\beta \approx 0.72$ ) required a panel regression with country and time fixed effects and clustered standard errors. By contrast, the simulator’s sliders apply deterministic arithmetic without confidence intervals or error terms.

Second, fixed elasticities ignore what economists call the Lucas Critique: the principle that behavioural parameters change when the policy regime changes. As I argued in my essay on Turkey’s central bank credibility, the effect of a 500-basis-point interest rate hike depends entirely on whether households and markets believe the central bank will stay the course. No fixed multiplier in a JavaScript array can capture a shift in institutional trust.

Third, real macroeconomic shocks rarely arrive one slider at a time. In historical data, supply shocks, exchange-rate depreciations, and interest-rate responses happen simultaneously. Separating cause from effect — the identification problem — requires formal econometric tools such as instrumental variables, panel GMM, or properly identified VAR models over long quarterly time series.

## Conclusion — From Transparent Illustration to Empirical Estimation

If a simple slider model has so many limitations, why include it on the site at all? Because for me as a student, writing the code forced me to translate verbal arguments from my essays — about import dependency, monetary transmission lags, and disinflation costs — into explicit mathematical relationships. The main lesson I take away from building the Macroeconomic Data Lab is that honesty about what a model cannot do is the foundation of serious research. The next

step for this portfolio is to replace hand-set weights with documented regressions estimated on a broader quarterly dataset — moving from assumptions made visible to hypotheses formally tested.

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